An Experimental Study of Cross-Lingual Context Isolation for Multilingual Answering

1 Abstract

Recent advancements in Retrieval-Augmented Generation (RAG) have enabled Large Language Models (LLMs) to access global knowledge across linguistic boundaries. However, the efficacy of small-scale multilingual LLMs in synthesising information from heterogeneous cross-lingual contexts remains under-explored. This work investigates whether multilingual context interference in small-scale Large Language Models can be mitigated through architectural decomposition strategies. Inspired by Recurrent Language Model (RLM) orchestration principles, we compare direct multilingual prompting against two alternative approaches: external translation-based context unification and isolated multilingual evidence extraction (inspired by Recurrent Language Model). [15] Experimental results on XQuAD using Vietnamese as the target language show that while decomposition strategies improve language fidelity and reduce instruction instability, they may also introduce semantic compression loss that reduces answer quality relative to direct multilingual prompting. [1]

2 Background and Importance

Large Language Models (LLMs) have demonstrated strong capabilities in natural language understanding and generation across multiple languages with the help of multi-head attention [13]. However, their effectiveness is heavily dependent on the availability and accessibility of knowledge across languages. In practice, English is often used as the primary interaction language with LLMs, despite the existence of high-quality resources in other languages such as Chinese.

This creates a potential knowledge accessibility gap, where valuable information in a non-English language may not be effectively utilised during inference. This issue is particularly critical for low-resource languages such as Chinese or Vietnamese, where local knowledge is limited and cross-lingual retrieval becomes essential.

3 Related Work

The challenge of multilingual context utilization in Large Language Models (LLMs) has recently been approached through the lens of mechanistic interpretability, focusing on how internal representations transition between languages.

3.1 Do Llamas Work in English? On the Latent Language of Multilingual Transformers

A primary concern in multilingual modelling is the disproportionate influence of English-centric training data. Wendler et al. [14] investigated the "internal pivot" hypothesis in models like Llama-2, discovering that non-English inputs are often transformed into an English-aligned "concept space" in the middle layers before being re-encoded into the target language at the output stage. This "thinking in English" phenomenon provides a potential explanation for our observed "English-augmented" performance peak; the model may find English context more computationally compatible with its internal reasoning logic than Vietnamese or Chinese.

3.2 Large Language Models Share Representations of Latent Grammatical Concepts Across Typologically Diverse Languages

Despite the dominance of English, evidence suggests that LLMs develop language-agnostic abstractions for structural concepts. Research by [4] demonstrates that typologically diverse languages share latent grammatical subspaces for features like tense and plurality. This indicates that while the "vocabulary" is distinct, the "logical folders" for grammar are unified. However, our findings suggest that in small-scale models like Qwen2-1.5B, the alignment of these subspaces may be insufficient to handle multi-source cross-lingual contexts, leading to the linguistic instability and "meaningless Vietnamese" outputs observed in our multilingual setup.

3.3 Separating Tongue from Thought: Activation Patching Reveals Language-Agnostic Concept Representations in Transformers C

The functional modularity of Transformers further clarifies the breakdown in multilingual reasoning. Recent work using activation patching [6] has shown a clear dissociation between language-agnostic "thought" heads and language-specific "tongue" layers. This architectural split suggests that while a model might correctly process a cross-lingual concept (Thought), the "Tongue" mechanism responsible for target-language constraints may fail under high cognitive load—such as when processing simultaneous English and Chinese contexts. Our observed "multilingual saturation" likely triggers a failure in these late-stage "Tongue" layers, causing the model to abandon Vietnamese instructions in favor of more robust English output.

4 Identified Limitation

Despite the competence of LLM, it still struggle to effectively utilize multilingual context, especially when multiple languages are provided simultaneously, leading to degraded or unstable generation of words. This will result in the model ignoring the non-English context and failing to integrate or use the different language contexts, thus producing degenerate outputs such as mindless repetition or hallucination.

5 Problem Definition

This work investigates the problem of multilingual context interference in small-scale multilingual Large Language Models (LLMs).

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Specifically, we study whether multilingual contextual decomposition can improve cross-lingual reasoning stability when the model is required to answer in a low-resource target language.

Given:

- an English context C_{en},
- a Chinese context C_{zh},
- a Vietnamese question Q_{vi},

the objective is to generate a Vietnamese answer A_{vi} that correctly integrates information from both contexts while preserving linguistic consistency.

Formally, the task can be expressed as:

$$A_{vi} = f(C_{en}, C_{zh}, Q_{vi})$$

where f represents the multilingual reasoning capability of the LLM.

Prior empirical observations from Task 2.2 demonstrated that directly concatenating multilingual contexts often leads to:

- instruction instability,
- hallucinated repetitions,
- unintended code-switching,
- degradation in semantic quality.

These failures suggest that multilingual contextual interference emerges when multiple linguistic representations compete within a single attention space. Recent studies further indicate that multilingual transformers internally rely on latent English-centric representations during reasoning [14]. Consequently, simultaneous multilingual context injection may overload the model's language alignment mechanisms.

The central hypothesis of this work is therefore:

> *Separating multilingual evidence processing into isolated intermediate reasoning stages can reduce multilingual interference and improve linguistic stability during multilingual question answering.*

To investigate this hypothesis, we evaluate three progressively structured multilingual reasoning pipelines:

- Direct multilingual context injection,
- Translation-based context unification,
- RLM-inspired isolated evidence extraction.

5.1 Research Hypothesis

We hypothesise that multilingual reasoning failures in small-scale Large Language Models are partially caused by contextual interference inside shared multilingual latent representations. When multiple heterogeneous languages are injected simultaneously into a single prompt, the model must jointly allocate attention across competing linguistic structures, token distributions, and semantic representations. This competition may overload the model's internal reasoning process, leading to instruction instability, semantic omission, hallucination, and uncontrolled code-switching.

Inspired by the decomposition principles proposed in Recursive Language Models (RLMs) [15], we propose that multilingual reasoning can be improved through contextual isolation and modular intermediate processing. Instead of forcing the model to reason over all multilingual evidence simultaneously, the reasoning process can be decomposed into smaller language-specific subtasks before final synthesis.

The central intuition is that separating multilingual evidence reduces representational interference during intermediate reasoning stages. Each sub-process can therefore focus on extracting semantically relevant information from a single linguistic context without competing multilingual activations. A final synthesis stage can then integrate the isolated evidence into a unified target-language response.

Furthermore, the recursive and modular design philosophy of RLM-inspired systems provides a potentially scalable framework for multilingual reasoning beyond fixed bilingual settings. Rather than designing architectures specifically for English-Chinese-Vietnamese interaction, recursive orchestration enables the addition of arbitrary contextual languages through repeated decomposition and synthesis stages. This suggests that context-isolated multilingual reasoning may generalise more effectively as the number of source languages increases.

Based on this hypothesis, we evaluate whether:

- external context normalization through translation,
- modular multilingual evidence isolation,
- and hierarchical synthesis strategies

can improve multilingual question answering performance relative to direct multilingual context injection.

6 Proposed Method

This work compares three multilingual reasoning architectures designed to answer Vietnamese questions using English and Chinese contextual evidence. All methods use the same base language model, dataset, and evaluation setup to ensure controlled comparison.

6.1 Direct Multilingual Context Injection

The first baseline directly concatenates English and Chinese contexts into a single prompt. The combined multilingual context is then provided to the language model together with the Vietnamese question.

The generation objective is defined as:

$$A_{vi} = f([C_{en}; C_{zh}], Q_{vi})$$

where $[;]$ denotes prompt concatenation.

This method represents the standard multilingual prompting strategy commonly used in multilingual Retrieval-Augmented Generation (RAG) systems. Its primary advantage is simplicity and preservation of complete contextual information without intermediate compression.

However, because all linguistic sources are injected simultaneously into a shared attention space, the model must internally resolve multilingual alignment during generation. Previous observations indicate that this frequently leads to multilingual saturation and unstable language switching.

6.2 External Context Translation

The second method introduces an explicit language normalization stage before reasoning. Instead of directly combining English and Chinese contexts, the Chinese context is first translated into English

using the Google Translate API through the `deep-translator` library. [5]

The transformed reasoning pipeline becomes:

$$C_{zh} \rightarrow C'_{en}$$

$$A_{vi} = f([C_{en}; C'_{en}], Q_{vi})$$

where C'_{en} represents the translated Chinese context.

This approach attempts to reduce multilingual representational conflict by forcing all retrieved evidence into a unified intermediate language prior to inference. The motivation is partially inspired by findings suggesting that multilingual transformers internally perform reasoning within an English-aligned latent space [14].

Unlike the direct multilingual approach, this method externalizes language alignment before the reasoning stage. Consequently, the model receives semantically unified contextual evidence while still generating the final answer in Vietnamese.

Nevertheless, this approach introduces an additional translation dependency that may propagate semantic compression errors or mistranslations into the final reasoning process.

6.3 RLM-Inspired Context Isolation

The third proposed architecture is inspired by Recursive Language Models (RLMs) [15]. Instead of forcing the model to jointly process multilingual evidence in a single prompt, the architecture decomposes multilingual reasoning into isolated intermediate stages.

The system operates through three sequential phases:

(1) Independent evidence extraction from English context by creating a separate instance of the main LLM model
(2) Independent evidence extraction from Chinese context by creating another separate instance of the main LLM model
(3) Final Vietnamese synthesis from extracted evidence

The architecture can be formalized as:

$$E_{en} = f_{sub}(C_{en}, Q_{vi})$$

$$E_{zh} = f_{sub}(C_{zh}, Q_{vi})$$

$$A_{vi} = f_{root}(E_{en}, E_{zh}, Q_{vi})$$

where:

- f_{sub} represents isolated sub-LLM evidence extraction,
- f_{root} represents final multilingual synthesis.

Unlike the original RLM framework which dynamically generates executable programs, our implementation adopts a simplified orchestration strategy using isolated sub-LLM calls without code generation. [15]

The key design principle is context isolation. Each sub-LLM processes only one language source independently, preventing direct multilingual interference during intermediate reasoning. [15] The final synthesis stage then combines already-processed semantic evidence rather than raw multilingual contexts.

This architecture is motivated by recent interpretability findings suggesting that multilingual transformers separate language-agnostic reasoning mechanisms from language-specific generation layers [6]. By isolating contextual processing, the model may reduce cognitive overload within the multilingual alignment mechanism.

7 Experimental Setup

7.1 Implementation and Reproducibility

All experiments were implemented in Python using the HuggingFace Transformers framework [12, 13]. The complete implementation, evaluation scripts, and experiment configurations are publicly available in our GitHub repository:

Multilingual Recurrent Language Model - github

7.2 Dataset

Experiments were conducted using the XQuAD multilingual question answering benchmark [2]. XQuAD extends the SQuAD dataset into multiple languages through professional human translation while preserving semantic equivalence across languages.

For each sample:

- English contexts were obtained from the English partition,
- Chinese contexts were obtained from the Chinese partition,
- Vietnamese questions and reference answers were obtained from the Vietnamese partition.

A total of 200 validation samples were evaluated under identical experimental conditions across all methods.

7.3 Base Model

All methods were implemented using the Qwen2.5-3B-Instruct model [12]. Qwen2.5 is a multilingual decoder-only transformer architecture supporting cross-lingual generation and instruction-following tasks.

To accommodate limited GPU memory constraints, experiments were conducted using 4-bit quantization through BitsAndBytes quantization configuration. Inference was executed using HuggingFace Transformers with automatic device mapping.

7.4 Evaluation Metrics

Multiple complementary evaluation metrics were used to assess both semantic correctness and multilingual stability.

7.4.1 BERTScore. BERTScore [16] evaluates semantic similarity between generated answers and reference answers using contextual embeddings derived from transformer representations.

Unlike lexical overlap metrics, BERTScore captures semantic equivalence even when wording differs significantly.

7.4.2 BLEU. BLEU [10] measures n-gram overlap between generated outputs and reference answers. Although originally designed for machine translation evaluation, BLEU remains useful for assessing lexical consistency.

7.4.3 chrF. chrF [11] computes character-level F-scores between predictions and references. Character-based evaluation is particularly useful for morphologically rich or multilingual outputs.

7.4.4 Language Fidelity. Language fidelity measures whether the generated output remains entirely within the requested target language (Vietnamese). This metric is computed using automatic language detection.

7.4.5 Code-Switch Rate. Code-switch rate measures the proportion of generated tokens identified as belonging to languages other than Vietnamese. This metric quantifies multilingual instability and unintended language switching [7].

8 Results and Analysis

Table 1 presents the averaged evaluation results across 200 multilingual question answering samples.

Method	BERTScore	BLEU	chrF	LF	CSR
Direct Multilingual Context Injection	0.6773	4.599	27.20	1.00	0.257
RLM-Inspired Context Isolation	0.7105	9.225	33.03	0.945	0.318
Translate-Then-Generate	**0.7116**	**9.205**	**34.33**	0.950	0.314

Table 1: Experimental results across multilingual reasoning methods. LF denotes Language Fidelity, while CSR denotes Code-Switch Rate. Higher BERTScore, BLEU, chrF, and LF are better, while lower CSR is preferred.

8.1 Multilingual Decomposition Significantly Improves Semantic Performance

Both decomposition-based architectures substantially outperformed direct multilingual context injection across all semantic evaluation metrics.

Compared to Direct Multilingual Context Injection, the RLM-inspired approach improved:

- BERTScore from 0.6773 to 0.7105,
- BLEU from 4.599 to 9.225,
- chrF from 27.20 to 33.03.

Similarly, Translate-Then-Generate achieved the highest overall semantic performance with:

- BERTScore of 0.7116,
- BLEU of 9.205,
- chrF of 34.33.

These findings strongly support the central hypothesis of this work: multilingual reasoning quality degrades when heterogeneous multilingual contexts are injected simultaneously into a single reasoning process.

Direct multilingual prompting forces the model to jointly allocate attention across competing linguistic representations. In contrast, decomposition-based methods reduce multilingual interference by isolating intermediate reasoning stages.

The large BLEU improvement observed in both decomposition architectures further suggests that contextual isolation improves lexical and structural alignment with Vietnamese reference answers rather than merely improving semantic similarity.

8.2 RLM-Inspired Context Isolation Achieves Near-Translation Performance Without External Translation

One of the most significant findings in this study is that the RLM-inspired architecture achieved performance nearly identical to the Translate-Then-Generate pipeline despite not relying on any external translation system.

The difference between the two approaches is extremely small:

- BERTScore difference of only 0.0011,
- BLEU difference of only 0.02,
- chrF difference of approximately 1.3 points.

This result is particularly important because the RLM-inspired pipeline processes Chinese context natively inside isolated sub-LLM reasoning stages rather than translating all multilingual evidence into English beforehand.

The findings therefore suggest that multilingual reasoning failure may not originate from an inability to understand multiple languages themselves. Instead, the primary limitation may arise from contextual interference caused by simultaneous multilingual reasoning inside a shared latent space.

By isolating Chinese and English reasoning into separate intermediate processing stages, the RLM-inspired architecture appears capable of preserving multilingual understanding while reducing cross-lingual representational conflict.

Conceptually, this behaviour aligns with the modular decomposition philosophy proposed in Recursive Language Models (RLMs) [15], where complex reasoning tasks are recursively divided into smaller isolated sub-problems before final synthesis.

Importantly, unlike translation-based normalization, the RLM-inspired design does not require collapsing all multilingual evidence into a dominant pivot language such as English. This may provide a more scalable framework for future multilingual systems involving larger numbers of source languages.

8.3 Translation-Based Context Normalization Remains the Strongest Overall Method

Although the performance gap is small, Translate-Then-Generate still achieved the strongest overall semantic performance.

This result supports prior findings suggesting that multilingual transformers internally rely on English-aligned latent reasoning representations [14**?**]. Translating Chinese context into English before reasoning likely reduces the cognitive complexity associated with multilingual alignment during inference.

The translation-based approach additionally achieved slightly better chrF performance than the RLM-inspired architecture, suggesting improved character-level alignment with Vietnamese references.

However, the marginal improvement over the RLM-inspired architecture may not justify the dependency on an external translation system. Translation pipelines introduce:

- additional computational overhead,
- external system dependency,
- potential translation artifacts,
- and information compression risk.

The near-equivalent performance achieved by native multilingual decomposition therefore represents a promising result for multilingual reasoning architectures that avoid explicit translation entirely.

8.4 Direct Multilingual Context Injection Remains the Weakest Strategy

Direct multilingual prompting consistently produced the weakest semantic performance across all evaluated metrics despite achieving the highest language fidelity score.

This result empirically validates the limitation identified in Task 2.2: small-scale multilingual LLMs struggle to simultaneously integrate heterogeneous multilingual evidence within a single reasoning process.

Interestingly, Direct Multilingual Context Injection achieved:

- the best Language Fidelity score,
- the lowest Code-Switch Rate,
- but the weakest semantic reasoning quality.

This suggests that linguistic stability alone does not imply effective multilingual reasoning. The model may preserve output-language consistency while still failing to properly integrate multilingual contextual information.

This behaviour is consistent with prior mechanistic interpretability findings [6, 14], which suggest that multilingual transformers partially separate language-specific decoding behaviour from language-agnostic reasoning representations.

Under direct multilingual prompting, the model likely experiences competition between multilingual latent activations during attention allocation, weakening reasoning quality despite preserving surface-level language consistency.

8.5 Trade-Off Between Semantic Reasoning and Linguistic Stability

An important observation across all methods is the emerging trade-off between semantic reasoning quality and linguistic stability.

Direct multilingual prompting achieved the strongest language fidelity and lowest code-switching rate but produced the weakest semantic performance. Conversely, both decomposition-based methods substantially improved reasoning quality while slightly increasing multilingual instability during generation.

This suggests that mechanisms responsible for multilingual reasoning and target-language control may operate somewhat independently inside multilingual transformers.

Architectures that improve multilingual information integration may unintentionally relax language-specific decoding constraints during generation.

Nevertheless, the increase in code-switching observed in decomposition architectures remained relatively small compared to the large gains in semantic performance. Therefore, the overall trade-off appears favourable for multilingual reasoning quality.

8.6 Error Analysis

Three primary failure modes were observed across experiments:

- **Instruction Drift:** Some generations partially ignored Vietnamese-only output constraints and inserted English phrases.
- **Code-Switching:** Mixed Vietnamese-English outputs remained common in semantically strong responses.
- **Evidence Compression Loss:** Intermediate summarisation occasionally removed contextual details necessary for precise factual answering.

Interestingly, the RLM-inspired architecture reduced catastrophic multilingual collapse observed in Task 2.2 while preserving native multilingual processing capabilities. This suggests that contextual isolation itself may be more important than explicit translation for multilingual reasoning robustness.

8.7 Limitations

Several limitations remain in this study.

First, experiments were conducted using a relatively small language model (Qwen2.5-3B). Larger multilingual models may exhibit stronger multilingual alignment capabilities and reduced contextual interference.

Second, only English and Chinese contextual sources were evaluated. Additional language combinations may reveal different multilingual interaction patterns.

Third, the current RLM-inspired implementation performs static orchestration rather than adaptive recursive planning as proposed in Recursive Language Models [15]. A fully recursive planner may further improve multilingual decomposition quality.

Finally, the current language fidelity and code-switch metrics rely on token-level language detection heuristics, which may imperfectly classify shared technical vocabulary across languages.

9 Conclusion and Future Work

This work investigated multilingual context interference in small-scale multilingual Large Language Models using Vietnamese multilingual question answering.

Experimental results demonstrate that direct multilingual prompting preserves the strongest semantic performance but suffers from multilingual instability and unintended code-switching. In contrast, translation-based normalization and RLM-inspired context isolation improve language fidelity while slightly reducing semantic quality due to intermediate information compression.

These findings suggest that multilingual reasoning involves a fundamental trade-off between semantic completeness and linguistic stability.

Future work may explore:

- adaptive recursive orchestration strategies,
- different combined language contexts
- multilingual retrieval routing,
- larger multilingual foundation models,
- attention-level multilingual isolation mechanisms,
- memory-augmented multilingual reasoning architectures.

Overall, the results provide empirical evidence that multilingual context decomposition is a promising direction for improving multilingual reasoning robustness in low-resource language settings.

References

[1] Mikel Artetxe, Sebastian Ruder, and Dani Yogatama. 2019. On the cross-lingual transferability of monolingual representations. CoRR abs/1910.11856 (2019). arXiv:1910.11856

[2] Mikel Artetxe, Sebastian Ruder, and Dani Yogatama. 2020. Cross-lingual Question Answering at Scale. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL 2020). 8769–8786. doi:10.18653/v1/2020.acl-main.784

[3] Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, et al. 2023. Qwen Technical Report. arXiv preprint arXiv:2309.16609 (2023).

[4] Jannik Brinkmann, Chris Wendler, Christian Bartelt, and Aaron Mueller. 2025. Large Language Models Share Representations of Latent Grammatical Concepts Across Typologically Diverse Languages. arXiv preprint arXiv:2501.06346 (2025). https://arxiv.org/abs/2501.06346

[5] Nidhal Chnita. 2020. deep-translator: A flexible free and unlimited python tool to translate between different languages in a simple way using multiple translators. https://github.com/nidhaloff/deep-translator GitHub repository. Star count: 1.8k as of 2024.

[6] Clément Dumas, Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. 2024. Separating Tongue from Thought: Activation Patching Reveals Language-Agnostic Concept Representations in Transformers. arXiv preprint arXiv:2411.08745 (2024). https://arxiv.org/abs/2411.08745

[7] Gualberto Guzmán et al. 2017. Metrics for Modeling Code-Switching Across Corpora. In Proceedings of the 21st Conference on Computational Natural Language Learning (CoNLL 2017). 67–75.

[8] Dan Hendrycks et al. 2021. Activation Patching for Model Mechanism Analysis. In International Conference on Learning Representations (ICLR 2021). https://arxiv.org/abs/2103.15053

[9] Carsten Holtermann et al. 2024. Evaluating the Elementary Multilingual Capabilities of Large Language Models. In Findings of the Association for Computational Linguistics: ACL 2024. 4463–4482.

[10] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. BLEU: a Method for Automatic Evaluation of Machine Translation. In Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL 2002). Association for Computational Linguistics, Philadelphia, Pennsylvania, USA, 311–318. doi:10.3115/1073083.1073135

[11] Maja Popović. 2015. chrF: character n-gram F-score for automatic MT evaluation. In Proceedings of the Tenth Workshop on Statistical Machine Translation (WMT 2015). Association for Computational Linguistics, Lisbon, Portugal, 392–395. doi:10.18653/v1/W15-3049

[12] Qwen Team. 2024. Qwen2 Technical Report. Technical Report. Alibaba Cloud. https://arxiv.org/abs/2407.10671 Qwen2 models: Qwen2-0.5B, Qwen2-1.5B, Qwen2-7B, Qwen2-57B-A14B, Qwen2-72B.

[13] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention Is All You Need. In Advances in Neural Information Processing Systems, Vol. 30. 5998–6008.

[14] Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. 2024. Do Llamas Work in English? On the Latent Language of Multilingual Transformers. In Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Association for Computational Linguistics, Bangkok, Thailand, 15366–15394. doi:10.18653/v1/2024.acl-long.841

[15] Alex L. Zhang, Tim Kraska, and Omar Khattab. 2026. Recursive Language Models. arXiv:2512.24601 [cs.AI] https://arxiv.org/abs/2512.24601

[16] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. BERTScore: Evaluating Text Generation with BERT. In International Conference on Learning Representations. https://openreview.net/forum?id=SkeHuCVFDr